SortSmart

Midterm Progress Report

CPE 595 – Applied Machine Learning

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Recap: Problem & Solution

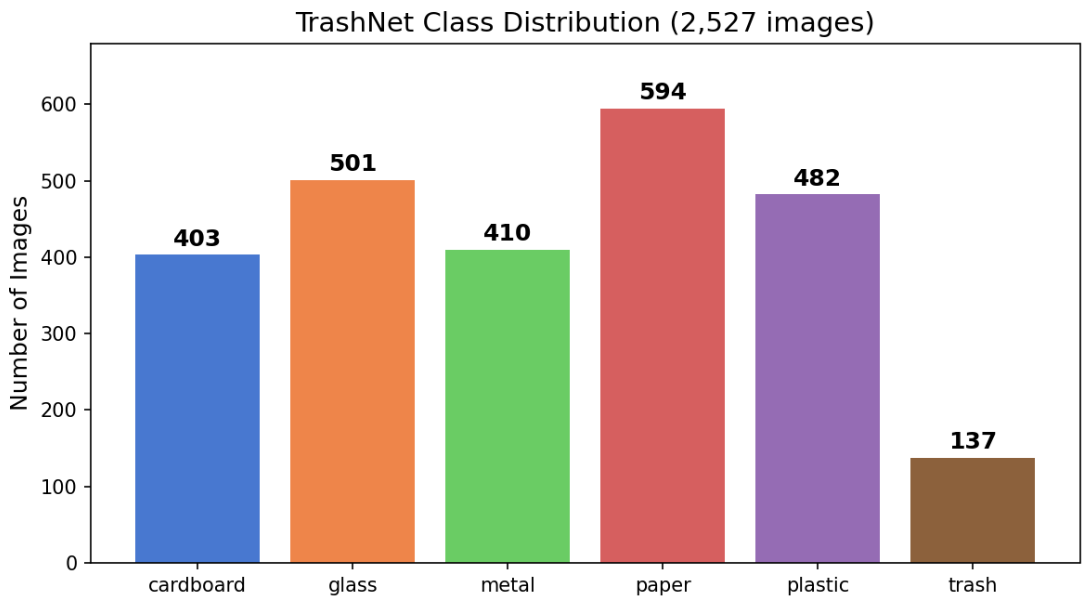
The Problem

- Recycling contamination costs MRFs millions annually
- ~17% average inbound contamination rate
- Wrong items cause entire loads to be landfilled
- US recycling rate stuck at ~32%

Our Solution

- Browser-based sorting assistant
- User snaps a photo of an item
- ML model classifies material + confidence
- App shows: recycle / trash / special handling
- Low-confidence fallback: ask user to retake or choose

Dataset & Augmentation



TrashNet Dataset

- 6 classes: cardboard, glass, metal, paper, plastic, trash
- 2,527 labeled images (resized to 128x128)
- Significant class imbalance ("trash" = only 137 images)
- Stratified 70/15/15 split before any augmentation

Augmentation (training set only)

- Random horizontal flip
- Brightness jitter (±40 intensity)
- Small rotation (±15°)
- Minority classes oversampled to 500 samples each
- No data leakage: augmentation applied only to train split

Augmentation Examples

- Each augmented image is a random combination of: horizontal flip, brightness shift (±40), and small rotation (±15°)
- Augmentation is applied only to training data after the train/val/test split (no data leakage)
- Minority classes (e.g. trash: 96 → 500 training samples) benefit the most from oversampling

Feature Engineering: HOG + HSV Color Histograms

HOG (Histogram of Oriented Gradients)

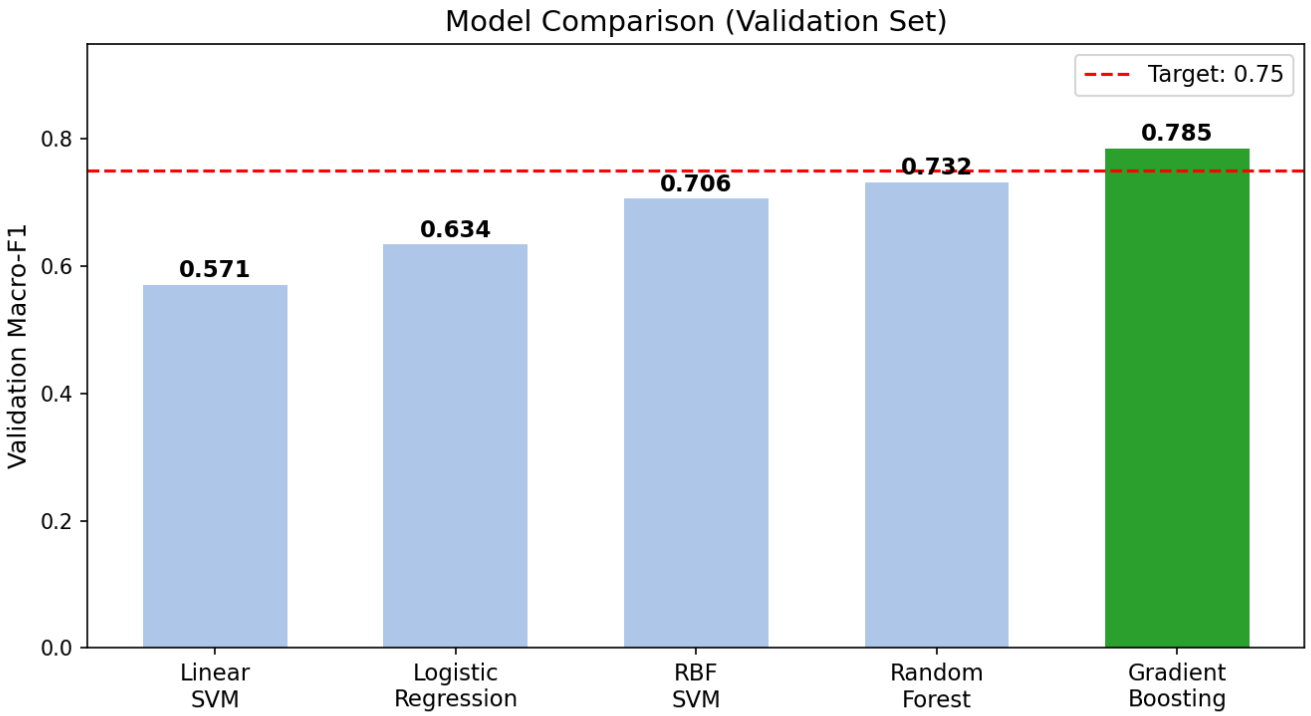
- Captures edge/texture structure of objects
- 9 orientations, 8x8 pixels per cell
- 2x2 cells per block, L2-Hys normalization
- Encodes shape information (flat paper vs curved bottle)
- 8,100 HOG features per image

HSV Color Histogram

- Captures color distribution of the item
- 32 bins per channel (H, S, V)
- L1-normalized for illumination invariance
- Distinguishes brown cardboard vs clear glass vs green plastic
- 96 color features per image

Combined feature vector: 8,196 dimensions per image | StandardScaler applied before training

Model Comparison



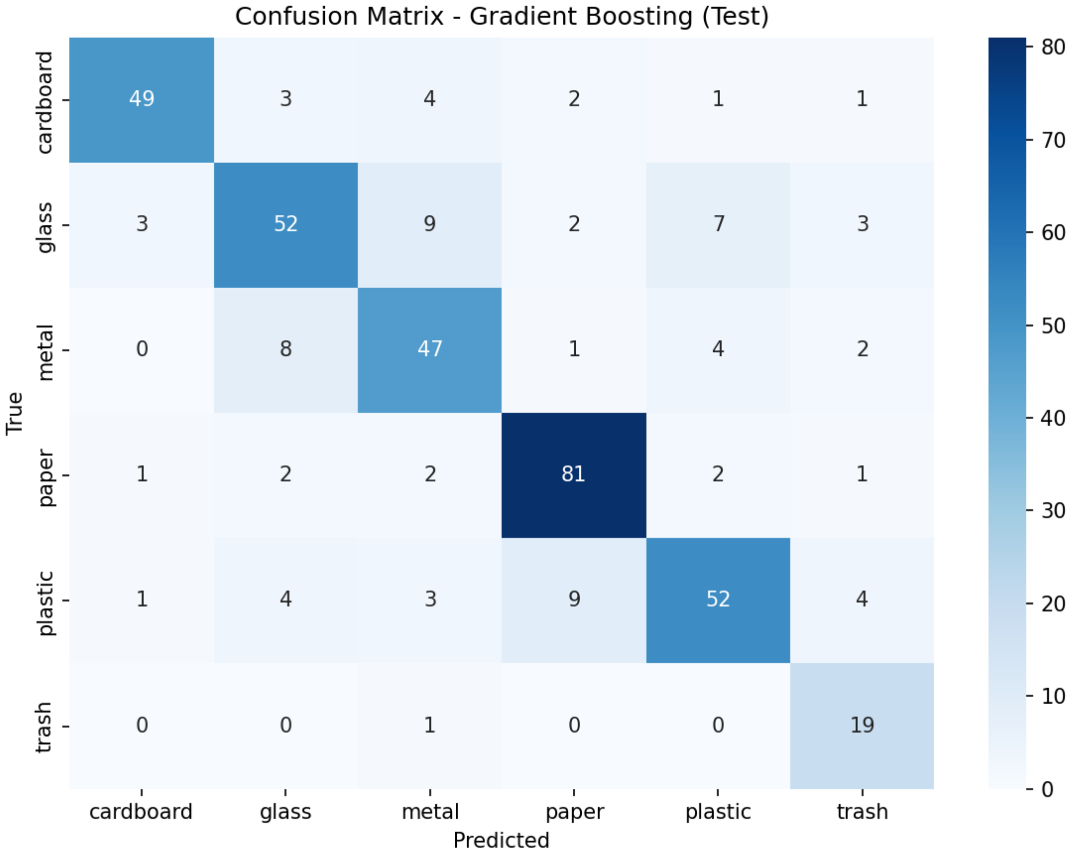
Key Takeaways

- 5 models trained on augmented training set (3,000 samples)
- All evaluated on clean validation set (379 samples)
- Gradient Boosting is the clear winner (F1 = 0.785)
- Random Forest close second (F1 = 0.732)
- Linear SVM struggles with high-dim features (F1 = 0.571)
- **Gradient Boosting exceeds 0.75 target**

Why Gradient Boosting?

- Handles high-dimensional features well
- Naturally captures non-linear feature interactions
- Built-in probability calibration

Test Set Results: Gradient Boosting



Overall Metrics

Accuracy

78.9%

Macro-F1

0.783

Median Latency

7.5 ms

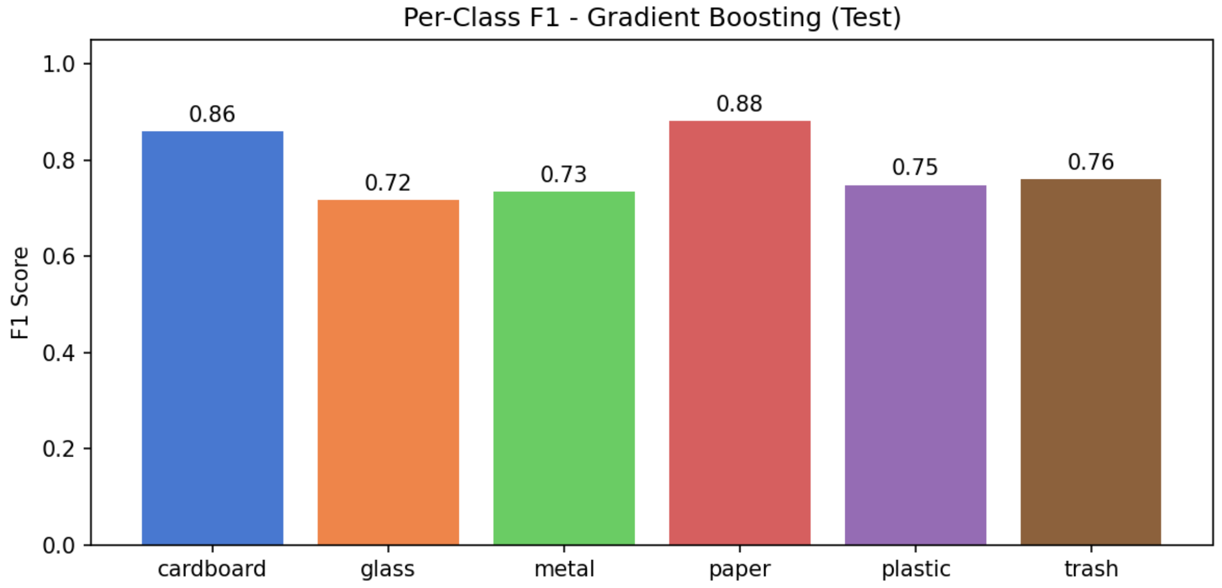
Test Samples

380

Confusion Analysis

- Glass most confused with metal (9 cases) - similar reflectivity
- Paper is the easiest class (91% recall)
- Trash has 95% recall but lower precision (63%) - model is aggressive

Per-Class Performance & Error Analysis



Strengths

- Paper (F1=0.88): distinct flat texture in HOG features
- Cardboard (F1=0.86): unique brown color + corrugated texture
- Trash (F1=0.76): huge improvement from 0.52 baseline via augmentation

Challenges

- Glass (F1=0.72): transparent objects vary in color, confused with metal
- Metal (F1=0.73): reflective surfaces create variable HOG patterns
- Plastic (F1=0.75): diverse shapes (bottles, bags, containers)

Improvement Ideas

- Additional texture features for glass vs metal
- More diverse training images for plastic subcategories

Success Criteria: All Targets Met

PASS **Macro-F1 >= 0.75**	0.783
PASS **Median Latency <= 1.5s**	7.5 ms
PASS **Confidence scores for predictions**	Calibrated probabilities
PASS **Low-confidence fallback UX**	Designed, implementation in progress

Iteration Path

Baseline (Linear SVM, 16x16 HOG, no aug) — **F1 = 0.62**

→ + Finer HOG (8x8) + augmentation — **F1 = 0.72**

→ + Gradient Boosting — **F1 = 0.78**

Next Steps & Remaining Work

Completed (W4-W6)

- Dataset loaded & explored
- Feature pipeline (HOG + HSV)
- 5 models trained & compared
- Augmentation for class balance
- Evaluation: all targets met

In Progress (W6-W9)

- Web app UI (Steven)
- Inference integration in browser
- Top-3 predictions + confidence bars
- k-nearest similar items display
- Low-confidence fallback flow

Remaining (W9-W14)

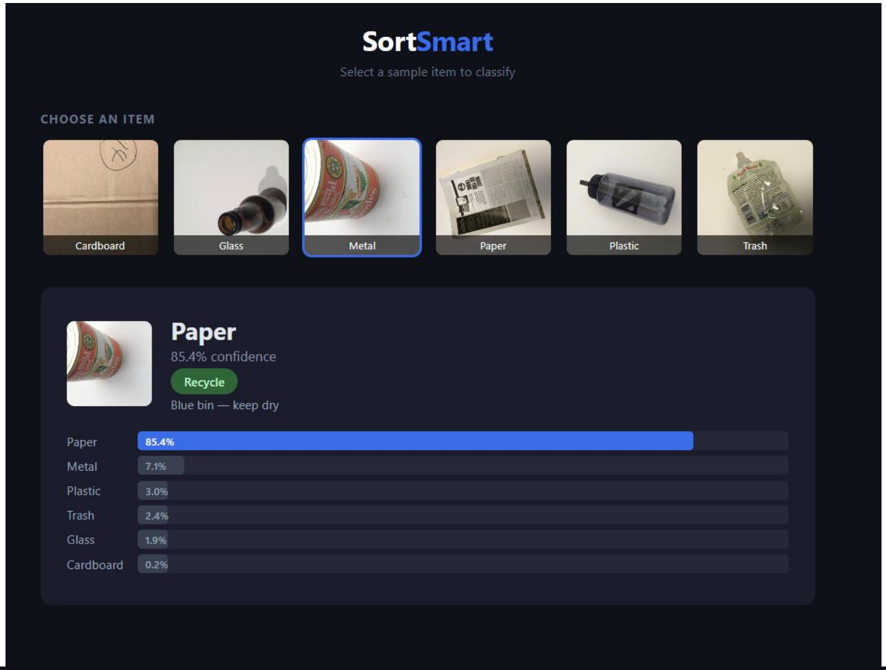
- End-to-end app deployment
- User testing & feedback
- Error analysis by condition
- Optional: team-collected images
- Final report & presentation

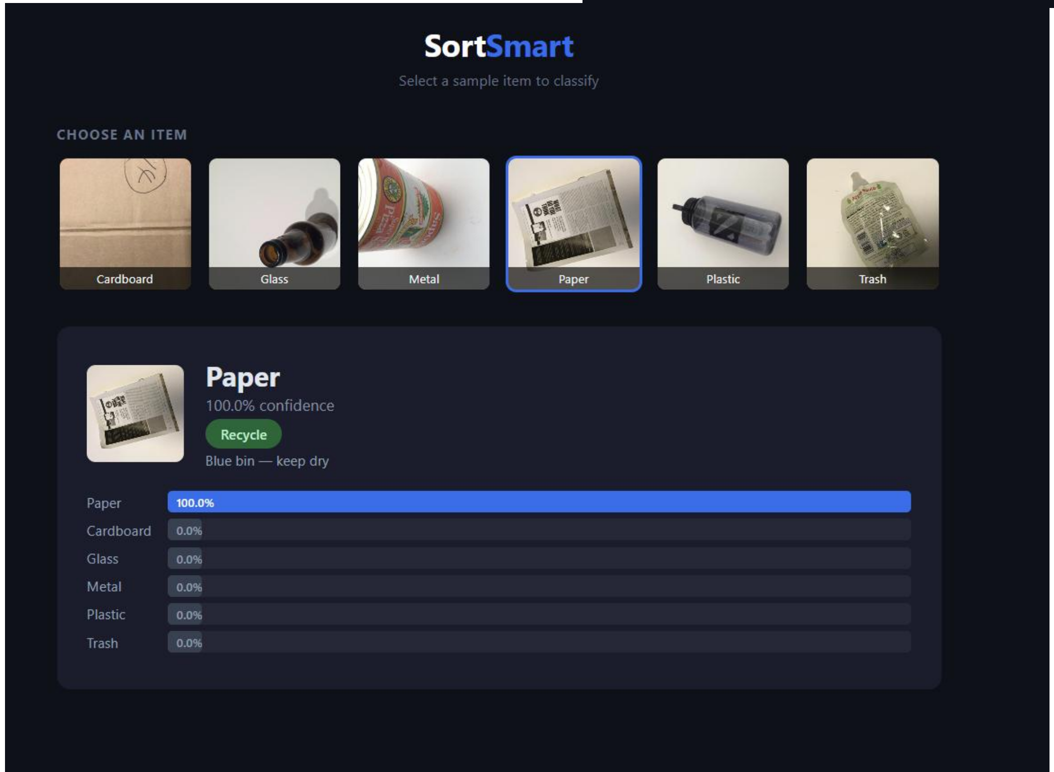
Team Roles

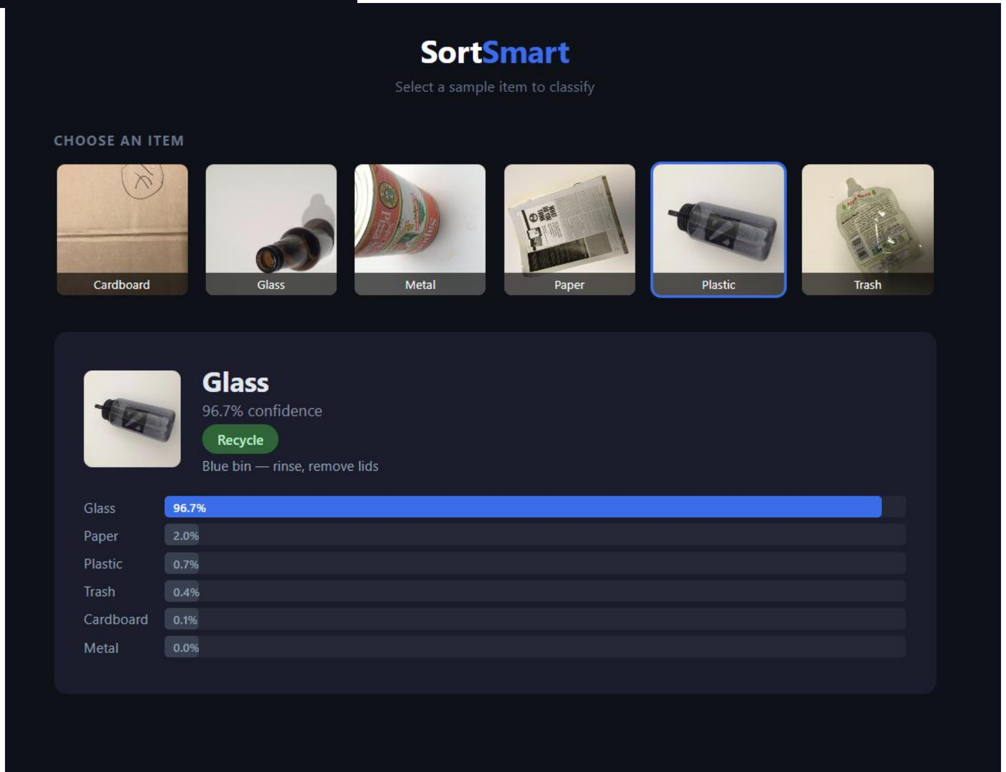
Melik: ML pipeline & metrics

Steven: App UI & deployment

Sam: Data, QA & edge cases


SortSmart
Select a sample item to classify
CHOOSE AN ITEM
Cardboard
Glass
Metal
Paper
Plastic
Trash
Paper
85.4% confidence
Recycle
Blue bin — keep dry
Paper 85.4%
Metal 7.1%
Plastic 3.0%
Trash 2.4%
Glass 1.9%
Cardboard 0.2%


SortSmart
Select a sample item to classify
CHOOSE AN ITEM
Cardboard
Glass
Metal
Paper
Plastic
Trash
Paper
100.0% confidence
Recycle
Blue bin — keep dry
Paper 100.0%
Cardboard 0.0%
Glass 0.0%
Metal 0.0%
Plastic 0.0%
Trash 0.0%


SortSmart
Select a sample item to classify
CHOOSE AN ITEM
Cardboard
Glass
Metal
Paper
Plastic
Trash
Glass
96.7% confidence
Recycle
Blue bin — rinse, remove lids
Glass 96.7%
Paper 2.0%
Plastic 0.7%
Trash 0.4%
Cardboard 0.1%
Metal 0.0%

Questions?